

Machine Learning Fundamentals

Machine Learning is a subset of Artificial Intelligence.

Types of Learning:

1. Supervised Learning: Classification, Regression
2. Unsupervised Learning: Clustering, Dimensionality Reduction
3. Reinforcement Learning: Learning through interaction

Linear Regression: Models linear relationship between variables.

Loss Function: Mean Squared Error (MSE)

Gradient Descent: Iterative optimization algorithm

Logistic Regression: Used for binary classification.

Sigmoid Function: Maps output to $[0,1]$ range

Neural Networks: Inspired by biological neurons.

Forward Propagation: Input $\rightarrow$ Hidden $\rightarrow$ Output

Backpropagation: Chain rule for gradient computation

Regularization: L1 (Lasso), L2 (Ridge) for preventing overfitting

Model Evaluation: Accuracy, Precision, Recall, F1-Score

Cross-Validation: K-fold validation technique

References: Bishop, Pattern Recognition and ML